

MANIKANTA SIRUMALLA

iOS Engineer | Swift, SwiftUI & Apple Watch | HealthKit | Applied AI

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F-1 CPT eligible | STEM OPT eligible after Dec 2026 (36 months) | Open to U.S. relocation

PROFESSIONAL SUMMARY

iOS Engineer with 5+ years of experience shipping native consumer apps in Swift, SwiftUI, and UIKit. Built and launched ZEALO, an Adaptive AI Fitness Intelligence platform across iPhone and Apple Watch with 1,300+ organic downloads and 1st Place at CBIC 2026. Experienced in HealthKit, Swift Concurrency, local-first architecture, AI/LLM integration, testing, performance, and App Store delivery.

TECHNICAL SKILLS

Languages: Swift, Python, SQL

iOS & Apple Platforms: iOS SDK, SwiftUI, UIKit, Swift Concurrency, async/await, multithreading, Combine, GCD, SwiftData, Core Data, HealthKit, WatchKit, WidgetKit, Live Activities, Dynamic Island, StoreKit 2, CoreML, Vision, NaturalLanguage, AVFoundation, MapKit, Swift Charts, PDFKit

Architecture & Data: MVVM, state management, protocol-oriented programming, dependency injection, modular feature architecture, local-first/offline-first systems, REST APIs, URLSession, Codable, JSON, iCloud sync, Firebase Analytics, Realtime Database, Crashlytics, App Check

AI & On-Device Intelligence: LLM integration, structured AI workflows, deterministic validation and fallbacks, on-device intent classification, privacy-first AI boundaries, CoreML

Quality & Tooling: XCTest, XCUITest, UI testing, Instruments, performance optimization, memory management and profiling, accessibility, Dynamic Type, VoiceOver, Xcode, Git/GitHub, Continuous Integration/Continuous Delivery (CI/CD), App Store Connect, Agile/Scrum

PROFESSIONAL EXPERIENCE

iOS Engineer & Founder | AutoClosure LLC

Feb 2025 – Present

Baltimore, MD

Products: [ZEALO on App Store](#) | zealofitness.com | [Tech Signal on App Store](#)

- Founded AutoClosure LLC and own end-to-end architecture, development, testing, App Store delivery, and production operations for two native iOS products, ZEALO and Tech Signal.
- Designed and launched ZEALO (Adaptive AI Fitness Intelligence), a privacy-first platform spanning iPhone, Apple Watch, widgets, Live Activities, and Dynamic Island, reaching 1,300+ organic downloads and winning 1st Place at CBIC 2026.
- Architected a hybrid deterministic + LLM system with structured outputs, validation gates, evidence-backed constraints, on-device intent routing, and offline fallbacks, reducing AI-coach execution from up to 9 model calls to a 4-call budget per question while improving privacy and failure handling.
- Built a 4- to 12-week adaptive training-plan engine with automated progression, constraint validation, retry-on-violation, and deterministic fallback, and validated generation through an 82/82 live-AI run, ~3,945-case test matrix, adversarial testing, and 6 XCUITest flows.
- Engineered HealthKit-based Recovery, Training Readiness, Strain, and Stress systems using personalized baselines, plus adaptive nutrition, menstrual-cycle-aware guidance, and Weekly Intelligence. Replaced an unsourced calorie model with 2024 Compendium MET values, correcting a 6-hour idle-workout estimate from 1,260 kcal to 53 kcal.
- Built the Apple Watch companion with live workout execution, heart-rate zones, motion-based rep detection, and effort scoring, and integrated StoreKit 2, iCloud sync, Firebase Analytics, Realtime Database, Crashlytics, and App Check for subscriptions, synchronization, telemetry, crash diagnostics, and app integrity.
- Expanded ZEALO beyond Apple Watch by integrating WHOOP, Oura, Fitbit, and Garmin, normalizing cross-device health, activity, sleep, heart-rate, and recovery signals into a unified intelligence layer for readiness, training, recovery, and personalized recommendations.
- Built and launched Tech Signal, aggregating 30+ developer sources with cache-first loading, personalized ranking, trend detection, and on-device summarization using Apple's Natural Language framework, with zero third-party runtime dependencies.

iOS Developer | ZetoStudio

Aug 2022 – Dec 2024

Hyderabad, India

- Built and shipped 10+ production iOS features in Swift and SwiftUI, owning work from product requirements through QA, App Store release, and post-release support.
- Designed 20+ modular feature components that separated networking, persistence, and view-model layers, improving testability, maintainability, and reuse across the codebase.
- Integrated REST APIs using URLSession, Codable, async/await, and Combine, and added offline-first caching to reduce unnecessary network requests and improve first-load reliability.
- Profiled complex screens with Instruments, memory analysis, image-pipeline optimization, and list virtualization, maintaining smooth 60 FPS interaction on older iOS devices while reducing UI stalls and memory pressure.
- Created reusable SwiftUI components and view-model patterns, wrote 100+ XCTest unit/UI tests, maintained CI workflows, and participated in code reviews to catch regressions before release.

iOS Developer | Grid Dynamics

Jul 2021 – Jul 2022

Hyderabad, India

- Built customer-facing iOS features for retail and e-commerce clients in Swift and UIKit, including product catalog, checkout, and account-management workflows.
- Integrated 15+ REST APIs, push-notification services, analytics SDKs, and reusable UI components to support scalable product features and consistent client experiences.
- Diagnosed and resolved 75+ pre-production and live defects across API connectivity, checkout flows, push notifications, and device-specific UI behavior while coordinating with product, QA, and backend teams.
- Migrated 15+ legacy UIKit components toward SwiftUI and Combine through modular refactoring, pair programming, architecture reviews, and incremental rollout.

PROJECTS

DermaFusion | On-Device Skin-Lesion Classification (Research Prototype)

[iOS App on GitHub](#) | [ML Pipeline on GitHub](#)

- Built a native SwiftUI iOS app for on-device skin-lesion detection, deploying a two-model EfficientNet-B4 ensemble as FP16 Core ML packages (34 MB each) so photos are analyzed entirely on iPhone with no cloud, sign-in, or analytics.
- Implemented camera capture and photo import, non-skin image rejection, Grad-CAM heatmap overlays, a malignant-risk gauge, and PDF report export via the iOS share sheet. Designed for educational research, not as a medical device.
- Trained the model on 61,694 dermoscopic and smartphone images unified from ISIC 2018/2019/2020 and PAD-UFES-20, using patient-grouped splits with zero patient/lesion overlap, Shades-of-Gray color constancy, and DullRazor hair removal.
- Optimized the ensemble with class-balanced focal loss, MixUp, EMA, Dirichlet-tuned weights, and per-class thresholds, reaching 0.830 malignant-vs-benign AUROC and 95.4% top-3 accuracy on a 5,279-image held-out test set, verified through automated clinical-readiness gates.

EDUCATION

M.S. Data Science | University of Maryland, Baltimore County (UMBC)

Jan 2025 – Dec 2026 (Expected)

Baltimore, Maryland · GPA: 3.97 / 4.00

B.Tech. Computer Science | Vidya Jyothi Institute of Technology

2017 – 2021

Hyderabad, India · GPA: 3.45 / 4.00

HONORS & RESEARCH

- **1st Place, Cangialosi Business Innovation Competition (CBIC) 2026:** Awarded a \$4,000 prize by bwTech@UMBC for ZEALO (formerly RepTrack Pro).
- **Apple WWDC 2026:** Selected for in-person attendance at Apple Park through an Apple Developer Program invitation.
- **First-author manuscript in preparation:** "ZEALO: A Hybrid AI-Deterministic Architecture for Personalized Fitness and Nutrition Plan Generation on iOS."